Muhammad Subhan Shahid

Aspiring AI / Machine Learning Engineer
Lahore, Pakistan

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SUMMARY

AI-focused undergraduate with hands-on experience in computer vision, machine learning, NLP, and model deployment. Skilled in convolutional neural networks, transformer architectures, reinforcement learning, and automating ML pipelines. Proven ability to design end-to-end AI systems and lead complex university projects. Actively seeking AI/ML internships.

EDUCATION

FAST National University of Computer and Emerging Sciences Aug 2023 – Present
Bachelors in Artificial Intelligence
Unique College, Lahore *2021 - 2023*
Intermediate in Computer Science (ICS)
Unique School, Lahore *2019 - 2021*
Matriculation in Computer Science

TECHNICAL SKILLS

- **AI / ML:** NLP, Reinforcement Learning, Neural Architecture Search, MLOps, LLMs, Prompt Engineering
- **Frameworks:** MLflow, Gradio, Streamlit, PyTorch, TensorFlow, Hugging Face, Scikit-learn
- **Programming:** Python, C, C++, SQL
- **Tools:** Git, GitHub, Linux, Makefiles, VS Code

PROJECTS

Agentic Dialogue Response Generation

- Transitioned standard Seq2Seq text generation into an Agentic AI framework to bridge conversational AI and autonomous tool execution.
- Fine-tuned a quantized Llama-3 (8B) model using LoRA/PEFT to enhance contextual dialogue and integrated it within a LangChain/LangGraph architecture to autonomously query external tools and vector databases.
- Utilized a dual-dataset approach, leveraging DailyDialog for multi-turn conversational fluency and ToolBench to train the model in agentic function-calling and JSON API structures.

Real-Time Market Movement Prediction System

- Architected a comprehensive MLOps pipeline for tracking and automating model training workflows.
- Utilized MLflow, DVC, Docker, Makefiles, and GitHub Actions to deploy the real-time prediction model effectively onto AWS infrastructure.

Conversational AI with RAG and Lightweight Agentic Tool Integration

- Developed a hybrid NLP conversational AI system that combines text generation, semantic retrieval, and agentic decision-making to seamlessly handle multi-turn dialogues.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline integrating a FAISS vector database to dynamically retrieve context from a custom knowledge base, significantly reducing hallucination and improving factual accuracy over baseline models.
- Built an intent-driven framework capable of evaluating user queries to selectively trigger and utilize external APIs when task-oriented answers are required.

Computer Vision for Classifying Pakistani Politicians

- Developed an end-to-end computer vision application to accurately classify images of political figures.
- Deployed the machine learning model utilizing FastAPI for the backend and Streamlit for an interactive, real-time user interface.

Fake Face (Deepfake) Detection System

- Developed a computer vision model utilizing Convolutional Neural Networks (CNNs) to accurately distinguish between real and AI-generated human faces.
- Preprocessed image datasets and optimized model weights to achieve high classification accuracy in identifying deepfake artifacts.

Fake News Detection System

- Built a classification pipeline using transformer models (BERT, RoBERTa, DistilBERT) to process and detect fabricated news.
- Fine-tuned models on custom datasets to generate high-accuracy classifications.

Text Summarization using Encoder-Decoder Models (BART, T5)

- Implemented abstractive text summarization using Transformer-based encoder-decoder architectures.
- Processed the arXiv dataset on Kaggle to generate concise and coherent summaries.

Phishing Email Detection & Deployment

- Trained and evaluated Logistic Regression, FNN, RNN, and LSTM models for accurately identifying phishing emails.
- Deployed the highest-performing NLP model through an interactive Gradio/Streamlit user interface.

PROFESSIONAL EXPERIENCE

Private Academic Tutor

Self-Employed | Lahore, Pakistan | 2021 – Present

- Managed a concurrent roster of 6 students over a 5-year period, demonstrating strong time management while simultaneously completing a rigorous BS Artificial Intelligence degree.
- Developed strong communication and mentoring skills by translating complex academic concepts into easily understandable lessons.
- Fostered a track record of reliability, adaptability, and independent problem-solving.

INTERESTS

- Agentic AI, Machine Learning Research, Autonomous Systems, Reinforcement Learning.